

symcoder parity-blindness demo

shadow event #0, parity = -1

1 About this document

This document is one half of a parity-blindness demonstration. A companion document, identical in structure but built from the parity-flipped event, exists alongside it. Their Phase 1 and Phase 2 encoded values should match exactly, despite the events lying in distinct $SO(3)$ orbits.

This run uses event with parity sign: This run uses event with parity sign: -1

Shadow event index: Shadow event index: 0 (from `symcoder.tests._parity_shadow_events.SHADOW_EVENT`)

Companion document: Companion document: `demo_shadow_event_parity+1.tex`

2 Setup

Particle types: electrons = $\{a, b, c\}$, muons = $\{p, q\}$

Group G : $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order = 12

Operations:

$|\mathbf{x}|$ (rank 1, symmetric)

$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)

$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):

a: $\mathbf{a} = (1.00, 1.00, 1.00)$

b: $\mathbf{b} = (1.00, 1.00, -0.00)$

c: $\mathbf{c} = (-1.00, 1.00, 1.00)$

p: $\mathbf{p} = (-0.00, -1.00, -0.00)$

q: $\mathbf{q} = (-0.00, 1.00, -0.00)$

3 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

8 FlavouredOperators in repS

3.1 FlavouredOperator: $|\mathbf{x}|$ flavour = $(1, 0)$

Canonical representative: $|\mathbf{a}|$

Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|a|, |b|, |c|$
Eval values: +1.7321, +1.4142, +1.7321
Encoded [3 reals]: +1.4142, +1.7321, +1.7321

3.2 FlavouredOperator: $|x|$ flavour = (0, 1)

Canonical representative: $|p|$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $|p|, |q|$
Eval values: +1.0000, +1.0000
Encoded [2 reals]: +1.0000, +1.0000

3.3 FlavouredOperator: $x \cdot y$ flavour = (2, 0)

Canonical representative: $a \cdot b$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $a \cdot b, a \cdot c, b \cdot c$
Eval values: +2.0000, +1.0000, +0.0000
Encoded [3 reals]: +0.0000, +1.0000, +2.0000

3.4 FlavouredOperator: $x \cdot y$ flavour = (1, 1)

Canonical representative: $a \cdot p$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $a \cdot p, a \cdot q, b \cdot p, b \cdot q, c \cdot p, c \cdot q$
Eval values: -1.0000, +1.0000, -1.0000, +1.0000, -1.0000, +1.0000
Encoded [6 reals]: -1.0000, -1.0000, -1.0000, +1.0000, +1.0000, +1.0000

3.5 FlavouredOperator: $x \cdot y$ flavour = (0, 2)

Canonical representative: $p \cdot q$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $p \cdot q$
Eval values: -1.0000
Encoded [1 reals]: -1.0000

3.6 FlavouredOperator: $\varepsilon(x, y, z)$ flavour = (3, 0)

Canonical representative: $\varepsilon(a, b, c)$
Encoder: HalfSortEncoder — stores $|\text{eval}|$ of sign= +1 representatives

Orbit atoms: $\varepsilon(a, b, c)$
Eval values: +2.0000
Encoded [1 reals]: +2.0000

3.7 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour = (2, 1)

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$

Eval values: -1.0000, +1.0000, +2.0000, -2.0000, +1.0000, -1.0000

Encoded [6 reals]: +1.0000, +1.0000, +1.0000, +1.0000, +2.0000, +2.0000

3.8 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour = (1, 2)

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$

Eval values: +0.0000, +0.0000, +0.0000

Encoded [3 reals]: +0.0000, +0.0000, +0.0000

4 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

4.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 2

Encoded values: +2.8284, +2.0000

Atom-pair orbit (multiset; ordering arbitrary):

(|**p**|, |**q**|)

(|**q**|, |**p**|)

Eval-pair multiset (u, v):

(+1.0000, +1.0000)

(+1.0000, +1.0000)

PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 1), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

(|**p**|, |**p**|)

(|**q**|, |**q**|)

Eval-pair multiset (u, v):

(+1.0000, +1.0000)

(+1.0000, +1.0000)

4.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +9.7566, -24.5959, +48.7832, -138.3837, +12.0136, -118.7878, -159.0918, +55.1918, -140.8499, +47.1918, -8.7710

Atom-pair orbit (multiset; ordering arbitrary):

(|**p**|, |**a**|)

(|**q**|, |**a**|)

(|**p**|, |**b**|)

(|**q**|, |**b**|)

(|**p**|, |**c**|)

$(|\mathbf{q}|, |\mathbf{c}|)$
 Eval-pair multiset (u, v) :
 $(+1.0000, +1.7321)$
 $(+1.0000, +1.7321)$
 $(+1.0000, +1.7321)$
 $(+1.0000, +1.7321)$
 $(+1.0000, +1.4142)$
 $(+1.0000, +1.4142)$

4.3 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 1), type=11,
 kind=ASSOC, encoded dim= 4

Encoded values: +2.0000, -2.0000, +0.0000, -2.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{p}|, \mathbf{p} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{p} \cdot \mathbf{q})$
 Eval-pair multiset (u, v) :
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$

4.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +18.0000, +0.0000, +32.0000, +0.0000, +36.0000, +0.0000, +24.0000, +0.0000, +8.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{p}|, \mathbf{a} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{a} \cdot \mathbf{p})$
 $(|\mathbf{p}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{p}|, \mathbf{c} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{c} \cdot \mathbf{p})$
 Eval-pair multiset (u, v) :
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, +1.0000)$
 $(+1.0000, +1.0000)$
 $(+1.0000, +1.0000)$

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11,
 kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +18.0000, +0.0000, +32.0000, +0.0000, +36.0000, +0.0000, +24.0000, +0.0000, +8.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{p}|, \mathbf{a} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{a} \cdot \mathbf{q})$
 $(|\mathbf{p}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{p}|, \mathbf{c} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{c} \cdot \mathbf{q})$

Eval-pair multiset (u, v) :

(+1.0000, -1.0000)
(+1.0000, -1.0000)
(+1.0000, -1.0000)
(+1.0000, +1.0000)
(+1.0000, +1.0000)
(+1.0000, +1.0000)

4.5 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +6.0000, +2.0000, +30.0000, -32.0000, +48.0000, -59.0000, +24.0000, -38.0000, -6.0000, -8.0000, -6.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\mathbf{a} \cdot \mathbf{b}$)
($|\mathbf{q}|$, $\mathbf{a} \cdot \mathbf{b}$)
($|\mathbf{p}|$, $\mathbf{a} \cdot \mathbf{c}$)
($|\mathbf{q}|$, $\mathbf{a} \cdot \mathbf{c}$)
($|\mathbf{p}|$, $\mathbf{b} \cdot \mathbf{c}$)
($|\mathbf{q}|$, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v) :

(+1.0000, +2.0000)
(+1.0000, +2.0000)
(+1.0000, +1.0000)
(+1.0000, +1.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)

4.6 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($|\mathbf{q}|$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($|\mathbf{p}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
($|\mathbf{q}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
($|\mathbf{p}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
($|\mathbf{q}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

Eval-pair multiset (u, v) :

(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, -0.0000)
(+1.0000, -0.0000)
(+1.0000, -0.0000)

4.7 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +12.0000, +78.0000, +340.0000, +1089.0000, +2664.0000, +5072.0000, +7536.0000, +8664.0000, +7520.0000, +4704.0000, +1920.0000, +400.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{p}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.0000, +2.0000)

(+1.0000, +2.0000)

(+1.0000, -2.0000)

(+1.0000, -2.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

... (4 more eval-pairs)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +12.0000, +78.0000, +340.0000, +1089.0000, +2664.0000, +5072.0000, +7536.0000, +8664.0000, +7520.0000, +4704.0000, +1920.0000, +400.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{p}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.0000, +2.0000)

(+1.0000, +2.0000)

(+1.0000, -2.0000)

(+1.0000, -2.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

(+1.0000, -1.0000)

... (4 more eval-pairs)

4.8 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 4

Encoded values: +4.0000, +14.0000, +20.0000, +25.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{p}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v) :

(+1.0000, +2.0000)

(+1.0000, +2.0000)

(+1.0000, -2.0000)

(+1.0000, -2.0000)

4.9 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +13.7980, +79.3939, +243.8684, +421.7673, +389.4439, +150.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $|\mathbf{b}|$)

($|\mathbf{a}|$, $|\mathbf{c}|$)

($|\mathbf{b}|$, $|\mathbf{a}|$)

($|\mathbf{b}|$, $|\mathbf{c}|$)

($|\mathbf{c}|$, $|\mathbf{a}|$)

($|\mathbf{c}|$, $|\mathbf{b}|$)

Eval-pair multiset (u, v) :

(+1.4142, +1.7321)

(+1.4142, +1.7321)

(+1.7321, +1.7321)

(+1.7321, +1.7321)

(+1.7321, +1.4142)

(+1.7321, +1.4142)

PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $|\mathbf{a}|$)

($|\mathbf{b}|$, $|\mathbf{b}|$)

($|\mathbf{c}|$, $|\mathbf{c}|$)

Eval-pair multiset (u, v) :

(+1.4142, +1.4142)

(+1.7321, +1.7321)

(+1.7321, +1.7321)

4.10 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +4.8783, -3.0000, +4.8990, -9.7566, -0.6357, -6.8990

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{p} \cdot \mathbf{q}$)

$(|\mathbf{b}|, \mathbf{p} \cdot \mathbf{q})$
 $(|\mathbf{c}|, \mathbf{p} \cdot \mathbf{q})$
 Eval-pair multiset (u, v) :
 $(+1.7321, -1.0000)$
 $(+1.7321, -1.0000)$
 $(+1.4142, -1.0000)$

4.11 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +19.5133, +0.0000, +180.3837, +0.0000, +1041.3172, +0.0000, +4173.3591, +0.0000, +12220.0778, +0.0000, +26792.4569, -0.0000, +44315.6341, -0.0000, +54903.0525, -0.0000, +49729.9824, -0.0000, +31305.6650, -0.0000, +12325.7542, +0.0000, +2304.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{a}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{a}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{a}|, \mathbf{c} \cdot \mathbf{p})$
 $(|\mathbf{a}|, \mathbf{c} \cdot \mathbf{q})$
 $(|\mathbf{b}|, \mathbf{a} \cdot \mathbf{p})$
 $(|\mathbf{b}|, \mathbf{a} \cdot \mathbf{q})$
 $(|\mathbf{b}|, \mathbf{c} \cdot \mathbf{p})$
 $(|\mathbf{b}|, \mathbf{c} \cdot \mathbf{q})$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(+1.4142, +1.0000)$
 $(+1.4142, +1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.7321, -1.0000)$
 $(+1.7321, -1.0000)$

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +9.7566, +0.0000, +42.5959, +0.0000, +105.0660, +0.0000, +154.3837, +0.0000, +128.3933, +0.0000, +48.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{a}|, \mathbf{a} \cdot \mathbf{p})$
 $(|\mathbf{a}|, \mathbf{a} \cdot \mathbf{q})$
 $(|\mathbf{b}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{b}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{c}|, \mathbf{c} \cdot \mathbf{p})$
 $(|\mathbf{c}|, \mathbf{c} \cdot \mathbf{q})$

Eval-pair multiset (u, v) :

$(+1.7321, -1.0000)$
 $(+1.7321, -1.0000)$
 $(+1.4142, -1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.7321, +1.0000)$
 $(+1.4142, +1.0000)$

4.12 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0,0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +4.8783, +3.0000, +5.8990, +9.7566, +0.7785, +7.8990

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{b} \cdot \mathbf{c}$)

($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{c}$)

($|\mathbf{c}|$, $\mathbf{a} \cdot \mathbf{b}$)

Eval-pair multiset (u, v) :

(+1.7321, +2.0000)

(+1.4142, +1.0000)

(+1.7321, +0.0000)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1,0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +9.7566, +6.0000, +26.5959, +48.7832, +1.2071, +146.3837, -96.9898, +198.7540, -140.8499, +114.7878, -60.9898, +17.6705

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{b}$)

($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{c}$)

($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{b}$)

($|\mathbf{b}|$, $\mathbf{b} \cdot \mathbf{c}$)

($|\mathbf{c}|$, $\mathbf{a} \cdot \mathbf{c}$)

($|\mathbf{c}|$, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v) :

(+1.7321, +2.0000)

(+1.4142, +2.0000)

(+1.7321, +1.0000)

(+1.7321, +1.0000)

(+1.7321, +0.0000)

(+1.4142, +0.0000)

4.13 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0,0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +19.5133, +174.3837, +943.7509, +3444.8245, +8934.3634, +16882.2855, +23417.7226, +23665.7018, +16992.6589, +8228.7183, +2412.9023, +324.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{c}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{a}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+1.7321, +0.0000)

(+1.7321, +0.0000)

(+1.7321, +0.0000)

(+1.7321, -0.0000)

(+1.7321, -0.0000)
 (+1.7321, -0.0000)
 (+1.7321, +0.0000)
 (+1.7321, +0.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=12,
 kind=ASSOC, encoded dim= 6
Encoded values: +9.7566, +39.5959, +85.5527, +103.7878, +67.0251, +18.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 Eval-pair multiset (u, v) :
 (+1.7321, +0.0000)
 (+1.7321, +0.0000)
 (+1.7321, +0.0000)
 (+1.7321, -0.0000)
 (+1.4142, -0.0000)
 (+1.4142, -0.0000)

4.14 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12,
 kind=ASSOC, encoded dim= 12
Encoded values: +19.5133, +186.3837, +1141.4262, +4963.5102, +16082.9329, +39708.0812, +75123.8548,
 +107957.5499, +114861.2004, +85907.3300, +40614.0886, +9216.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (+1.4142, -2.0000)
 (+1.4142, -2.0000)
 (+1.4142, +2.0000)
 (+1.4142, +2.0000)
 (+1.7321, +1.0000)
 (+1.7321, +1.0000)
 (+1.7321, +1.0000)
 (+1.7321, +1.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +19.5133, +0.0000, +186.3837, +0.0000, +1137.6121, +0.0000, +4907.0857, -
 0.0000, +15685.5403, -0.0000, +37980.0812, -0.0000, +70047.4790, +0.0000, +97562.9173, +0.0000,

+100076.0612, +0.0000, +71849.2564, +0.0000, +32509.9005, +0.0000, +7056.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{b}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{b}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.7321, -2.0000)

(+1.7321, -2.0000)

(+1.7321, +2.0000)

(+1.7321, +2.0000)

(+1.7321, -1.0000)

(+1.7321, -1.0000)

(+1.4142, -1.0000)

(+1.4142, -1.0000)

... (4 more eval-pairs)

4.15 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +9.7566, +51.5959, +163.6057, +342.1714, +429.5775, +294.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{a}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{b}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{c}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(+1.7321, -2.0000)

(+1.7321, -2.0000)

(+1.4142, -2.0000)

(+1.7321, +2.0000)

(+1.7321, +2.0000)

(+1.4142, +2.0000)

4.16 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 2), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{p} \cdot \mathbf{q}$)

Eval-pair multiset (u, v):

(-1.0000, -1.0000)

4.17 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: -6.0000, +0.0000, +18.0000, +0.0000, -32.0000, +0.0000, +36.0000, +0.0000, -24.0000, +0.0000, +8.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{p}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{q}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{q}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{q}$)

Eval-pair multiset (u, v):

(-1.0000, +1.0000)

(-1.0000, +1.0000)

(-1.0000, +1.0000)

(-1.0000, -1.0000)

(-1.0000, -1.0000)

(-1.0000, -1.0000)

4.18 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: -3.0000, +3.0000, +1.0000, -6.0000, +1.0000, +3.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{b}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{c}$)

($\mathbf{p} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v):

(-1.0000, +2.0000)

(-1.0000, +1.0000)

(-1.0000, +0.0000)

4.19 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 2), type=12, kind=ASSOC, encoded dim= 6

Encoded values: -6.0000, +15.0000, -20.0000, +15.0000, -6.0000, +1.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

Eval-pair multiset (u, v):

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, -0.0000)

(-1.0000, -0.0000)

(-1.0000, -0.0000)

4.20 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: -12.0000, +78.0000, -340.0000, +1089.0000, -2664.0000, +5072.0000, -7536.0000, +8664.0000, -7520.0000, +4704.0000, -1920.0000, +400.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(-1.0000, -2.0000)

(-1.0000, -2.0000)

(-1.0000, +2.0000)

(-1.0000, +2.0000)

(-1.0000, +1.0000)

(-1.0000, +1.0000)

(-1.0000, +1.0000)

(-1.0000, +1.0000)

... (4 more eval-pairs)

4.21 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 2

Encoded values: -2.0000, +5.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(-1.0000, -2.0000)

(-1.0000, +2.0000)

4.22 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +12.0000, +0.0000, +60.0000, +0.0000, +160.0000, +0.0000, +240.0000, +0.0000, +192.0000, +0.0000, +64.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$)

($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$)

($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$)

($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$)

$(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{q})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{p})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{c} \cdot \mathbf{q})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{p})$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(-1.0000, +1.0000)$
 $(-1.0000, +1.0000)$
 ... (4 more eval-pairs)
 PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11_sigma,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, -12.0000, +0.0000, +60.0000, +0.0000, -160.0000, +0.0000, +240.0000,
 +0.0000, -192.0000, +0.0000, +64.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{p})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{q})$
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{c} \cdot \mathbf{p})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{q})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{p})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{q})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{c} \cdot \mathbf{p})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{q})$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 $(-1.0000, -1.0000)$
 $(-1.0000, -1.0000)$
 $(-1.0000, -1.0000)$
 $(-1.0000, -1.0000)$
 $(-1.0000, -1.0000)$
 $(-1.0000, -1.0000)$
 $(+1.0000, +1.0000)$
 $(+1.0000, +1.0000)$
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma,
 kind=ASSOC, encoded dim= 6
Encoded values: +0.0000, +6.0000, +0.0000, +12.0000, +0.0000, +8.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{q})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{p})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{q})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{p})$
 $(\mathbf{c} \cdot \mathbf{p}, \mathbf{c} \cdot \mathbf{q})$
 $(\mathbf{c} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{p})$
 Eval-pair multiset (u, v) :
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$
 $(+1.0000, -1.0000)$

(-1.0000, +1.0000)
 (-1.0000, +1.0000)
 (-1.0000, +1.0000)
 PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 1),
 type=null_self, kind=NULL_SELF, encoded dim= 0
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{p}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{q}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{p}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{q}$)
 ($\mathbf{c} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{p}$)
 ($\mathbf{c} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{q}$)
 Eval-pair multiset (u, v):
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 (+1.0000, +1.0000)
 (+1.0000, +1.0000)
 (+1.0000, +1.0000)

4.23 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +6.0000, -16.0000, +0.0000, +0.0000, -24.0000, +25.0000, +0.0000,
 +0.0000, +18.0000, -10.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{c}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{c}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{c}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{c}$)
 ($\mathbf{c} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{c} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{b}$)
 Eval-pair multiset (u, v):
 (+1.0000, +2.0000)
 (-1.0000, +2.0000)
 (+1.0000, +1.0000)
 (+1.0000, +0.0000)
 (-1.0000, +1.0000)
 (-1.0000, +0.0000)
 PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +12.0000, -68.0000, +0.0000, +0.0000, -240.0000, +594.0000, +0.0000,
 +0.0000, +1104.0000, -1612.0000, +0.0000, +0.0000, -1896.0000, +1809.0000, +0.0000, +0.0000, +1380.0000,
 -824.0000, +0.0000, +0.0000, -360.0000, +100.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{c}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{c}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{c}$)

$(\mathbf{b} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{c})$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (+1.0000, +2.0000)
 (+1.0000, +2.0000)
 (-1.0000, +2.0000)
 (-1.0000, +2.0000)
 (-1.0000, +1.0000)
 (-1.0000, +1.0000)
 (-1.0000, +0.0000)
 (-1.0000, +0.0000)
 ... (4 more eval-pairs)

4.24 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 1), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, -6.0000, +0.0000, +0.0000, +0.0000, +15.0000, +0.0000, +0.0000,
 +0.0000, -20.0000, +0.0000, +0.0000, +0.0000, +15.0000, +0.0000, +0.0000, +0.0000, -6.0000, +0.0000,
 +0.0000, +0.0000, +1.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (-1.0000, +0.0000)
 (-1.0000, +0.0000)
 (-1.0000, +0.0000)
 (-1.0000, -0.0000)
 (-1.0000, -0.0000)
 (-1.0000, -0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 1), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +0.0000, -3.0000, +0.0000, +0.0000, +0.0000, +3.0000, +0.0000, +0.0000,
 +0.0000, -1.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 Eval-pair multiset (u, v) :
 (-1.0000, +0.0000)

(-1.0000, +0.0000)
 (-1.0000, +0.0000)
 (+1.0000, -0.0000)
 (+1.0000, -0.0000)
 (+1.0000, -0.0000)

4.25 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +6.0000, +0.0000, +33.0000, +0.0000, +48.0000, +0.0000, +216.0000, +0.0000, +96.0000, +0.0000, +400.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{c} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{c} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{p}$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(-1.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 (+1.0000, +2.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 (+1.0000, -1.0000)
 (+1.0000, -1.0000)

... (4 more eval-pairs)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +6.0000, +0.0000, +33.0000, +0.0000, +48.0000, +0.0000, +216.0000, +0.0000, +96.0000, +0.0000, +400.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{c} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{c} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{p}$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(-1.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 (+1.0000, +2.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)

(+1.0000, -1.0000)
 (+1.0000, -1.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1,0), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +6.0000, +0.0000, +0.0000, +0.0000, +33.0000, +0.0000, +0.0000,
 +0.0000, +48.0000, +0.0000, +0.0000, +0.0000, +216.0000, +0.0000, +0.0000, +0.0000, +96.0000,
 +0.0000, +0.0000, +0.0000, +400.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (-1.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 (+1.0000, +2.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 (+1.0000, -1.0000)
 (+1.0000, -1.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1,1), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +6.0000, +0.0000, +0.0000, +0.0000, +33.0000, +0.0000, +0.0000,
 +0.0000, +48.0000, +0.0000, +0.0000, +0.0000, +216.0000, +0.0000, +0.0000, +0.0000, +96.0000,
 +0.0000, +0.0000, +0.0000, +400.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (-1.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 (+1.0000, +2.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 (+1.0000, -1.0000)
 (+1.0000, -1.0000)
 ... (4 more eval-pairs)

4.26 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +18.0000, +0.0000, +183.0000, +0.0000, +1116.0000, +0.0000, +4575.0000, +0.0000, +11250.0000, +0.0000, +15625.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{c} \cdot \mathbf{p}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{c} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a} \cdot \mathbf{p}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.0000, -2.0000)
 (+1.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, -2.0000)
 (-1.0000, -2.0000)
 (-1.0000, -2.0000)
 (+1.0000, +2.0000)
 (+1.0000, +2.0000)

... (4 more eval-pairs)

4.27 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +8.4853, +32.0000, +67.8823, +85.0000, +59.3970, +20.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$)
 ($\mathbf{a} \cdot \mathbf{c}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{a} \cdot \mathbf{b}$, $\mathbf{b} \cdot \mathbf{c}$)
 ($\mathbf{b} \cdot \mathbf{c}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{c}$)
 ($\mathbf{b} \cdot \mathbf{c}$, $\mathbf{a} \cdot \mathbf{c}$)

Eval-pair multiset (u, v):

(+0.0000, +2.0000)
 (+1.0000, +2.0000)
 (+2.0000, +1.0000)
 (+2.0000, +0.0000)
 (+0.0000, +1.0000)
 (+1.0000, +0.0000)

PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{b}$)
 ($\mathbf{a} \cdot \mathbf{c}$, $\mathbf{a} \cdot \mathbf{c}$)

$(\mathbf{b} \cdot \mathbf{c}, \mathbf{b} \cdot \mathbf{c})$
 Eval-pair multiset (u, v) :
 (+0.0000, +0.0000)
 (+1.0000, +1.0000)
 (+2.0000, +2.0000)

4.28 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=12,
 kind=ASSOC, encoded dim= 6

Encoded values: +6.0000, +13.0000, +12.0000, +4.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{b}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{b}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

Eval-pair multiset (u, v) :

(+2.0000, +0.0000)
 (+2.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, -0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=12,
 kind=ASSOC, encoded dim= 12

Encoded values: +12.0000, +62.0000, +180.0000, +321.0000, +360.0000, +248.0000, +96.0000, +16.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{b}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{b}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{b}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+2.0000, +0.0000)
 (+2.0000, +0.0000)
 (+2.0000, +0.0000)
 (+2.0000, -0.0000)
 (+1.0000, -0.0000)
 (+1.0000, -0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)

... (4 more eval-pairs)

4.29 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +12.0000, +0.0000, +74.0000, +0.0000, +300.0000, +0.0000, +897.0000, +0.0000, +2088.0000, +0.0000, +3868.0000, +0.0000, +5688.0000, +0.0000, +6612.0000, +0.0000, +6000.0000, +0.0000, +4208.0000, +0.0000, +2112.0000, +0.0000, +640.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($\mathbf{a} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($\mathbf{b} \cdot \mathbf{c}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{b} \cdot \mathbf{c}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+2.0000, -2.0000)

(+0.0000, -2.0000)

(+2.0000, +2.0000)

(+0.0000, +2.0000)

(+2.0000, +1.0000)

(+2.0000, -1.0000)

(+0.0000, -1.0000)

(+1.0000, -1.0000)

... (4 more eval-pairs)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (2, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +12.0000, +74.0000, +300.0000, +879.0000, +1944.0000, +3340.0000, +4536.0000, +4959.0000, +4380.0000, +3050.0000, +1500.0000, +625.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($\mathbf{a} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($\mathbf{b} \cdot \mathbf{c}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($\mathbf{b} \cdot \mathbf{c}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($\mathbf{a} \cdot \mathbf{b}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($\mathbf{a} \cdot \mathbf{b}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.0000, -2.0000)

(+1.0000, -2.0000)

(+2.0000, -1.0000)

(+2.0000, -1.0000)

(+2.0000, +1.0000)

(+2.0000, +1.0000)

(+1.0000, +2.0000)

(+1.0000, +2.0000)

... (4 more eval-pairs)

4.30 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (2, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +6.0000, +25.0000, +60.0000, +124.0000, +144.0000, +160.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{b}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
($\mathbf{a} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
($\mathbf{b} \cdot \mathbf{c}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
($\mathbf{a} \cdot \mathbf{b}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
($\mathbf{a} \cdot \mathbf{c}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
($\mathbf{b} \cdot \mathbf{c}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(+2.0000, +2.0000)
(+0.0000, +2.0000)
(+1.0000, +2.0000)
(+2.0000, -2.0000)
(+0.0000, -2.0000)
(+1.0000, -2.0000)

4.31 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (0, 2), type=neg_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
($\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
($\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
... (4 more eval-pairs)

PairFlavour overlap=(1, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 2), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
($\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
($-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

$(-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 Eval-pair multiset (u, v) :
 $(+0.0000, +0.0000)$
 $(+0.0000, +0.0000)$
 $(+0.0000, +0.0000)$
 $(-0.0000, -0.0000)$
 $(-0.0000, -0.0000)$
 $(-0.0000, -0.0000)$

4.32 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +12.0000, +0.0000, +54.0000, +0.0000, +116.0000, +0.0000, +129.0000, +0.0000, +72.0000, +0.0000, +16.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $\dots$ (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(+0.0000, +2.0000)$
 $(+0.0000, +2.0000)$
 $(+0.0000, -2.0000)$
 $(-0.0000, -2.0000)$
 $(-0.0000, -1.0000)$
 $(-0.0000, -1.0000)$
 $(+0.0000, -1.0000)$
 $(+0.0000, -1.0000)$
 $\dots$ (4 more eval-pairs)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 1), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +12.0000, +0.0000, +0.0000, +0.0000, +54.0000, +0.0000, +0.0000, +0.0000, +116.0000, +0.0000, +0.0000, +0.0000, +129.0000, +0.0000, +0.0000, +0.0000, +72.0000, +0.0000, +0.0000, +0.0000, +16.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $\dots$ (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(+0.0000, +2.0000)$
 $(+0.0000, +2.0000)$

(+0.0000, -2.0000)
 (-0.0000, -2.0000)
 (-0.0000, -1.0000)
 (-0.0000, -1.0000)
 (+0.0000, -1.0000)
 (+0.0000, -1.0000)
 ... (4 more eval-pairs)

4.33 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=22 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=22, kind=ASSOC, encoded dim= 6

Encoded values: -24.0000, +240.0000, -1280.0000, +3840.0000, -6144.0000, +4096.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+0.0000, +2.0000)
 (+0.0000, +2.0000)
 (+0.0000, -2.0000)
 (-0.0000, -2.0000)
 (-0.0000, +2.0000)
 (-0.0000, +2.0000)
 (+0.0000, -2.0000)
 (+0.0000, -2.0000)
 ... (4 more eval-pairs)

4.34 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +0.0000, +4.0000, +0.0000, +0.0000, +10.0000, +0.0000, +0.0000, +0.0000, +0.0000, +56.0000, +0.0000, +0.0000, +569.0000, +0.0000, +0.0000, +0.0000, +0.0000, +2500.0000, +0.0000, +0.0000, -2500.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ... (4 more atom-pairs)

Eval-pair multiset (u, v):

```

(-2.0000, -1.0000)
(-1.0000, -2.0000)
(-2.0000, +1.0000)
(-1.0000, +2.0000)
(+1.0000, +2.0000)
(+2.0000, +1.0000)
(-1.0000, +1.0000)
(-1.0000, +1.0000)
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 PairFlavour overlap= (1, 1), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +0.0000, -4.0000, +0.0000, +0.0000, +10.0000, +0.0000, +0.0000,
+0.0000, +0.0000, -56.0000, +0.0000, +0.0000, +569.0000, +0.0000, +0.0000, +0.0000, +0.0000, -
2500.0000, +0.0000, +0.0000, -2500.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
( $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ ,  $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ )
( $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ ,  $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ )
( $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ ,  $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ )
( $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ ,  $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ ,  $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ ,  $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$ )
( $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ )
( $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ )
... (4 more atom-pairs)
Eval-pair multiset ( $u, v$ ):
(-2.0000, +1.0000)
(-1.0000, +2.0000)
(-2.0000, -1.0000)
(-1.0000, -2.0000)
(+1.0000, -2.0000)
(+2.0000, -1.0000)
(-1.0000, -1.0000)
(-1.0000, -1.0000)
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=neg_sigma kind=ASSOC dim=6 PairFlavour overlap= (2, 0), type=neg_sigma,
kind=ASSOC, encoded dim= 6
Encoded values: -24.0000, -216.0000, +928.0000, +2064.0000, -2304.0000, -1024.0000
Atom-pair orbit (multiset; ordering arbitrary):
( $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ ,  $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ )
( $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ ,  $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ )
( $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ ,  $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ )
( $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ ,  $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ )
( $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$ ,  $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$ )
... (4 more atom-pairs)
Eval-pair multiset ( $u, v$ ):
(+2.0000, -2.0000)
(-2.0000, +2.0000)
(+2.0000, -2.0000)
(-2.0000, +2.0000)
(+1.0000, -1.0000)

```

(-1.0000, +1.0000)
 (+1.0000, -1.0000)
 (-1.0000, +1.0000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 1),
 type=null_self, kind=NULL_SELF, encoded dim= 0
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v):
 (+1.0000, +1.0000)
 (+1.0000, +1.0000)
 (+1.0000, +1.0000)
 (+1.0000, +1.0000)
 (+2.0000, +2.0000)
 (+2.0000, +2.0000)
 (-1.0000, -1.0000)
 (-1.0000, -1.0000)
 ... (4 more eval-pairs)

4.35 **Block:** $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (**block**)

PairFlavour overlap=(2, 0) type=neg kind=ASSOC dim=12 **PairFlavour** overlap= (2, 0), type=neg,
 kind=ASSOC, encoded dim= 12
Encoded values: -12.0000, +0.0000, +150.0000, +0.0000, -1068.0000, +0.0000, +6129.0000, +0.0000,
 -19200.0000, +0.0000, +40000.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v):
 (+2.0000, -2.0000)
 (-2.0000, +2.0000)
 (+2.0000, +2.0000)
 (-2.0000, -2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 (+1.0000, -2.0000)
 (-1.0000, +2.0000)
 ... (4 more eval-pairs)

4.36 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(3, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (3, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

Eval-pair multiset (u, v) :

$(+2.0000, +2.0000)$

$(-2.0000, -2.0000)$

5 Alignment Decoder (recovers the G-orbit of repS evaluations)

—repS—: $|repS| = 8$

—G—: $|G| = 12$

Decoded vectors: decoded column vectors: 12

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table**
(rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
$ \mathbf{a} $	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142
$ \mathbf{p} $	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
$\mathbf{a} \cdot \mathbf{b}$	+1.0000	+1.0000	+0.0000	+0.0000	+2.0000	+2.0000	+0.0000	+0.0000	+0.0000	+2.0000	+2.0000	+2.0000
$\mathbf{a} \cdot \mathbf{p}$	-1.0000	+1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	-2.0000	-2.0000	-1.0000	-1.0000	-1.0000	-1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	+0.0000	-0.0000	-0.0000	-0.0000	-0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000

Ground truth (direct evaluation of g·repS on event): **Ground truth** (direct evaluation of $g \cdot \text{repS}$ on event):

Multiset comparison (decoded vs ground truth): **Multiset comparison:**

MISMATCH — decoded multiset differs from ground truth

Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**

Mymag(a): $|\mathbf{a}|$: max err = $3.18e - 01$

Mymag(p): $|\mathbf{p}|$: max err = $0.00e + 00$

dot(a,b): $\mathbf{a} \cdot \mathbf{b}$: max err = $0.00e + 00$

dot(a,p): $\mathbf{a} \cdot \mathbf{p}$: max err = $2.00e + 00$

Table 2: Ground truth alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$ \mathbf{a} $	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.7321	+1.7321
$ \mathbf{p} $	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
$\mathbf{a} \cdot \mathbf{b}$	+2.0000	+2.0000	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+1.0000	+1.0000
$\mathbf{a} \cdot \mathbf{p}$	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	+2.0000	+2.0000	-2.0000	-2.0000	-2.0000	-2.0000	+2.0000	+2.0000	+2.0000	+2.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	-1.0000	+1.0000	+2.0000	-2.0000	+1.0000	-1.0000	+1.0000	-1.0000	-2.0000	+2.0000
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000

dot(p,q): $\mathbf{p} \cdot \mathbf{q}$: max err = $0.00e+00$

eps(a,b,c): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$: max err = $4.00e+00$

eps(a,b,p): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err = $0.00e+00$

eps(a,p,q): $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$: max err = $0.00e+00$

6 Summary

Phase 1 encoded length: Phase 1 encoded length: 25 reals

Phase 2 encoded length: Phase 2 encoded length: 624 reals

Total encoded length: Total encoded length: 649 reals

repS size: $|\text{repS}| = 8$ atoms

—G—: $|G| = 12$ group elements

Decoded vectors: decoded alignment vectors: 12

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_shadow_event_parity_-1.tex
 Compile with: pdflatex demo_shadow_event_parity_-1.tex